

Technology Stack

Date	02 October 2026
Team ID	SWTID-2026-7991
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side React + Vite JavaScript, HTML, CSS	React + Vite	Responsive student interface, reusable components and fast development.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	FastAPI + Python	FastAPI backend for clean REST endpoints and efficient API handling.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	SQLite	Lightweight structured storage for MVP learning/session data.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Render	Cloud deployment option for hosting the FastAPI backend with simple scaling.
5	Version Control & CI/CD	Git + GitHub	Version control, team collaboration and tracking of project changes.

	<i>(e.g., GitHub, GitLab, Jenkins)</i>	Google Gemini API	Generates AI-powered answers, simplified explanations, summaries, quizzes and learning recommendations.
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>		